Project Report

Students Grade System

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Abstract
The Student Grade System is a terminal-based program that allows users to load, view, add, modify, and delete student grade records stored in a text file. The project ensures modularity by breaking the whole program into multiple packages and modules.

Problem Statement

Implement a well-structured Python program that enables an instructor to maintain their students' grades. The grade information is kept in a text file where the '#' symbol separates the ID from the name and from the grades, i.e., each line has 2 '#' symbols. The grades of each student are separated by one or more blanks. Each line of the text file contains a unique student ID, the student's first and last names, followed by test grades. No fixed number for the number of tests.

Example:

91007# Ahmad Said# 50.0 78.5 73.2
91004# Hassan Khan# 45.5 36.7 88.5
91003# Suleiman Wasim# 72.6 66.4 56.7

Furthermore, the program needs to have functionality to add, delete, view, and modify test grades.

Solution Approach

The program converts the database into a nested Python dictionary. **Student ID** is the key to the outer dictionary, and the other information is the values. The inner dictionary consists of **'name'** and **'grades'** as key and their respective value pairs.

The structure is shown below:

```
students = {
   "student_id1": {
         "name": "Student Name 1",
         "grades": ["grade1", "grade2", "grade3", ...]
   }
}
```

Its structure is used to ease the search mechanism of the program.

To extract data from the database, the program uses a combination of .split() and .strip() string functions and stores them in the **students** dictionary.

The formatting is shown below.

```
student_id, student_name, student_grades = line.split("#")

students[student_id.strip()] = {
         "name" : student_name.strip(),
         "grades" : student_grades.split()
         }
```

Using this dictionary across all the functions, the program ensures all the required functionality of the program.

Project Structure and Modularity

Initially, the project was created in a single main.py file. To improve maintainability, it has been divided into multiple packages and modules.

```
STUDENTS-GRADE-SYSTEM
  core_functions
    __pycache__
    __init__.py
    add_grade.py
    add_student.py
    delete_student.py
    grade_all_student.py
    grade_average.py
    grade_one_student.py
    modify_grades.py
  database
    __pycache__
    data
      grades.txt
    __init__.py
    load_students.py
    save_students.py
  exceptions
    __pycache__
    __init__.py
    student_id_error.py
  files
    CSE_101_Project_Spring_2026.pdf
    project-report.pdf
  ui
    __pycache__
    __init__.py
    enter.py
    header.py
    menu.py
  utility
    __pycache__
    __init__.py
    database_print.py
    test_count.py
  main.py
```

Core Functions Description

1. *load_students():*
   Reads "FILENAME" file line by line and formats the data in the **students** dictionary.

2. *grade_all_student():*
   Displays all students with all test scores in a table.

3. *grade_one_student():*
   Takes input (Student ID) from the user and displays the specific student with all test scores.

4. *grade_average():*
   Displays all students with the average of all test scores in a table.

5. *modify_grades():*
   Takes input (Student ID, No of test which grade needs to be modified, and the new grade) from the user and changes the grade in the dictionary. After that, this function displays the specific student with all test scores before modification and after modification.

6. *add_grade():*
   Allows users to input grades for a new test for every student in the dictionary.

7. *add_students():*
   Takes input for a new student (ID and Name) along with their grade and appends it to the dictionary.

8. *delete_student():*
   Removes a student by ID using dict.pop(student_id).

9. *save_students():*
   Converts the dictionary into the format of the file and writes the files with the data from the dictionary.

   file.write(f"{student_id}#{name}#{grades_string}\n")

Other Functions Description

1. *database_print():*
   Takes input in either "test" mode (all scores) or "avg" mode (average). Then, print data from our dictionary in a formatted way.

2. *header():*
   A reusable block that prints a dynamic table header of the database depending on the input format {"test" (all test numbers) or "avg" (average)}

3. *test_count():*
   Returns the number of tests by checking the first student's grades list, and returns 0 if there are no students.

4. *menu():*
   Prints the main menu with all 8 options

5. *enter():*
   A reusable block that takes input with "Press Enter to continue..."

1. *Exception: StudentIDError:*
   A custom exception class that handles duplicate ID on add and missing ID on delete.

Challenges Faced and How I Solved Them

1. *How to manage the dataset:*
   There was 2 possible ways to choose from:
   a. Using Lists
   b. Using Dictionary

   Using lists would give me a 3-dimensional matrix. To access student grades, I would have needed to specify them using a matrix, something like students[1][2][0]. This is a very efficient way, but it's not a developer-friendly way, as students[1][2][0] are difficult to track and prone to error. Instead, I chose a nested dictionary containing internal lists. This format provides a more readable and robust way to manage the data.
2. *Addition of the new grade to the add_grade() function:*
   I have created a temporary dictionary,

```
collected = {
        "student_id1": "85.0",
        "student_id2": "90.0"
    }
```

   Then I appended the temporary dictionary to the main **students** dictionary.

3. *Keeping the display formatted and aligned for varying name/grade lengths:*
   I have used f-string alignment specifiers in database_print() and header(), and calculated the character lengths to give it the perfect formatted alignment.

4. *Locating the data file after modulation, regardless of where the script is run:*
   When running the program from different directories, the program couldn't find the database by passing “grades.txt” or "data/grades.txt" in open(), no matter what I do. The open() function can only open a file if that file remains in the same directory as the module calling it. That’s why I needed to construct an absolute path relative to the file using os.path.join(os.path.dirname(__file__), "data", FILE_NAME)

ONE UNSOLVED CHALLENGE

Gaps in the grade history lead to indexing errors; when there are missing intermediate tests, new scores are incorrectly shifted to fill the empty slots during the save/load process, which results in mislabeled data.

Conclusion

In this project, a console-based Student Grade Management System was built, which ensures all the functionality, starting from loading the database from an existing file and saving it after making changes. It can show grades of all students as well as their average grades, even if I want to get grades for only one particular student. Furthermore, it can add, modify, and delete records from the database as well through an interactive menu-driven interface

Appendix

menu():

```
------------------------------------------------------------
              Student Grade Management System
------------------------------------------------------------
1. Display Grade Info for all students
2. Display Grade Info for a particular student
3. Display tests average for all students
4. Modify a particular test grade for a particular student
5. Add test grades for a particular test for all students
6. Add a new Student
7. Delete a student
8. Save and Exit
------------------------------------------------------------

Please select your choice: 
```

grade_all_student():

```
Please select your choice: 1

StudentID  Student Name         Test1 Test2 Test3 Test4 Test5

    91012  Galib               45.0  87.0
    91007  Ahmad Said          50.0  78.5  73.2  32.0  78.0
    91004  Hassan Khan         45.5  36.7  88.5  76.0  98.0
    91003  Suleiman Wasim      72.6  66.4  56.7  89.0  58.0
    91002  Majed Sameer        60.0  78.4  45.6  8.0   89.0
    91006  Muhammad Adel       85.5  69.8  44.5  76.0  78.0
    91005  Muhsim Zuheir       70.0  62.1  95.4  54.0  89.0
    91010  Fahim Hossain       98.0  88.0  90.0  76.0  59.0
    91011  Prasun Paul         89.0  92.0  95.0  98.0  68.0

Press Enter key to continue . . .
```

grade_one_student():

```
Please select your choice: 2

Enter studentID: 90
Error: Invalid student ID

Press Enter key to continue . . .

------------------------------------------------------------
              Student Grade Management System
------------------------------------------------------------
1. Display Grade Info for all students
2. Display Grade Info for a particular student
3. Display tests average for all students
4. Modify a particular test grade for a particular student
5. Add test grades for a particular test for all students
6. Add a new Student
7. Delete a student
8. Save and Exit
------------------------------------------------------------

Please select your choice: 2

Enter studentID: 91010
StudentID  Student Name         Test1 Test2 Test3 Test4 Test5
    91010  Fahim Hossain        98.0  88.0  90.0  76.0  59.0

Press Enter key to continue . . .
```

grade_average():

```
Please select your choice: 3

StudentID  Student Name         Average

    91012  Galib                66.0
    91007  Ahmad Said           62.3
    91004  Hassan Khan          68.9
    91003  Suleiman Wasim       68.5
    91002  Majed Sameer         56.2
    91006  Muhammad Adel        70.8
    91005  Muhsim Zuheir        74.1
    91010  Fahim Hossain        82.2
    91011  Prasun Paul          88.4

Press Enter key to continue . . .
```

modify_grades():

```
Please select your choice: 4

Enter studentID: 91001
Error: Invalid student ID

Press Enter key to continue . . .

------------------------------------------------------------
              Student Grade Management System
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1. Display Grade Info for all students
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8. Save and Exit
------------------------------------------------------------

Please select your choice: 4

Enter studentID: 91004
Please enter quiz number to modify: 0
Error: Please enter a number in between 1 and 5
Please enter quiz number to modify: 1
Please enter new quiz 1 grade: 67
Before grade modification:     91004  Hassan Khan          45.5  36.7  88.5  76.0  98.0
After grade modification:      91004  Hassan Khan          67.0  36.7  88.5  76.0  98.0

Press Enter key to continue . . .
```

add_grade():

```
Please select your choice: 5

Please enter test grades for Test#6

Please enter grade for student : 91012
89
Please enter grade for student : 91007
78
Please enter grade for student : 91004
102
Error: Grade must be between 0 and 100. Please try again.
Please enter grade for student : 91004
45
Please enter grade for student : 91003
-57
Error: Grade must be between 0 and 100. Please try again.
Please enter grade for student : 91003
57
Please enter grade for student : 91002
==
Error: Please enter a numeric grade.
Please enter grade for student : 91002
90
Please enter grade for student : 91006
65
Please enter grade for student : 91005
89
Please enter grade for student : 91010
90
Please enter grade for student : 91011
78

Press Enter key to continue . . .
```

```
Please select your choice: 1

StudentID  Student Name         Test1 Test2 Test3 Test4 Test5 Test6

    91012  Galib                45.0  87.0                    89.0
    91007  Ahmad Said           50.0  78.5  73.2  32.0  78.0  78.0
    91004  Hassan Khan          67.0  36.7  88.5  76.0  98.0  45.0
    91003  Suleiman Wasim       72.6  66.4  56.7  89.0  58.0  57.0
    91002  Majed Sameer         60.0  78.4  45.6  8.0   89.0  90.0
    91006  Muhammad Adel        85.5  69.8  44.5  76.0  78.0  65.0
    91005  Muhsim Zuheir        70.0  62.1  95.4  54.0  89.0  89.0
    91010  Fahim Hossain        98.0  88.0  90.0  76.0  59.0  90.0
    91011  Prasun Paul          89.0  92.0  95.0  98.0  68.0  78.0

Press Enter key to continue . . .
```

add_students():

```
Please select your choice: 6

Enter studentID: 91010
Student ID already exists

Press Enter key to continue . . .

------------------------------------------------------------
              Student Grade Management System
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1. Display Grade Info for all students
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6. Add a new Student
7. Delete a student
8. Save and Exit
------------------------------------------------------------

Please select your choice: 6

Enter studentID: 91001
Enter Student Name: Mahid
Enter grade for test#1: 90
Enter grade for test#2: 89
Enter grade for test#3: 68
Enter grade for test#4: 30
Enter grade for test#5: 89
Enter grade for test#6: 58

Press Enter key to continue . . .
```

delete_student():

```
Please select your choice: 7

Enter studentID: 91020
Student ID doesn't exist

Press Enter key to continue . . .

------------------------------------------------------------
              Student Grade Management System
------------------------------------------------------------
1. Display Grade Info for all students
2. Display Grade Info for a particular student
3. Display tests average for all students
4. Modify a particular test grade for a particular student
5. Add test grades for a particular test for all students
6. Add a new Student
7. Delete a student
8. Save and Exit
------------------------------------------------------------

Please select your choice: 7

Enter studentID: 91002

Press Enter key to continue . . .
```

```
Please select your choice: 1

StudentID  Student Name         Test1 Test2 Test3 Test4 Test5 Test6

    91012  Galib                45.0  87.0                    89.0
    91007  Ahmad Said           50.0  78.5  73.2  32.0  78.0  78.0
    91004  Hassan Khan          67.0  36.7  88.5  76.0  98.0  45.0
    91003  Suleiman Wasim       72.6  66.4  56.7  89.0  58.0  57.0
    91006  Muhammad Adel        85.5  69.8  44.5  76.0  78.0  65.0
    91005  Muhsim Zuheir        70.0  62.1  95.4  54.0  89.0  89.0
    91010  Fahim Hossain        98.0  88.0  90.0  76.0  59.0  90.0
    91011  Prasun Paul          89.0  92.0  95.0  98.0  68.0  78.0
    91001  Mahid                90.0  89.0  68.0  30.0  89.0  58.0

Press Enter key to continue . . .
```

save_students():

```
Please select your choice: 8

PS D:\code\codequest-completed-projects-archive\students-grade-system\students-grade-system> python main.py

------------------------------------------------------------
              Student Grade Management System
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1. Display Grade Info for all students
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4. Modify a particular test grade for a particular student
5. Add test grades for a particular test for all students
6. Add a new Student
7. Delete a student
8. Save and Exit
------------------------------------------------------------

Please select your choice: 1

StudentID  Student Name         Test1 Test2 Test3 Test4 Test5 Test6

    91012  Galib               45.0  87.0  89.0
    91007  Ahmad Said          50.0  78.5  73.2  32.0  78.0  78.0
    91004  Hassan Khan         67.0  36.7  88.5  76.0  98.0  45.0
    91003  Suleiman Wasim      72.6  66.4  56.7  89.0  58.0  57.0
    91006  Muhammad Adel       85.5  69.8  44.5  76.0  78.0  65.0
    91005  Muhsim Zuheir       70.0  62.1  95.4  54.0  89.0  89.0
    91010  Fahim Hossain       98.0  88.0  90.0  76.0  59.0  90.0
    91011  Prasun Paul         89.0  92.0  95.0  98.0  68.0  78.0
    91001  Mahid               90.0  89.0  68.0  30.0  89.0  58.0

Press Enter key to continue . . .
```